ECS 170 Artificial Intelligence
Course Project Stage 3 Report

Team information: To be completed by students.

Section 1: Task Description

We trained convolutional neural networks (CNNs) to recognize handwritten digits, faces, and color objects. The instructor provided separate training and test partitions for MNIST, ORL, and CIFAR-10. For each dataset, we trained a baseline CNN and two architecture variants. We selected checkpoints using validation loss and evaluated them on the supplied test partition.

Section 2: Model Description

All three models use convolution and ReLU blocks followed by 2x2 pooling. Adaptive average pooling reduces the final feature map to 4x4, and a fully connected hidden layer produces raw class scores. Cross-entropy loss takes those scores directly, without a softmax layer. The baselines differ in input channels, convolution widths, and output classes: MNIST uses 1->32->64 channels and 10 outputs; ORL uses 1->16->32->64 channels and 40 outputs; CIFAR uses 3->32->64->128 channels and 10 outputs. Each baseline convolution has a 3x3 kernel, padding 1, and stride 1. Max pooling follows each block, and the hidden layer has 128 units. The architecture diagram shows this shared pipeline.

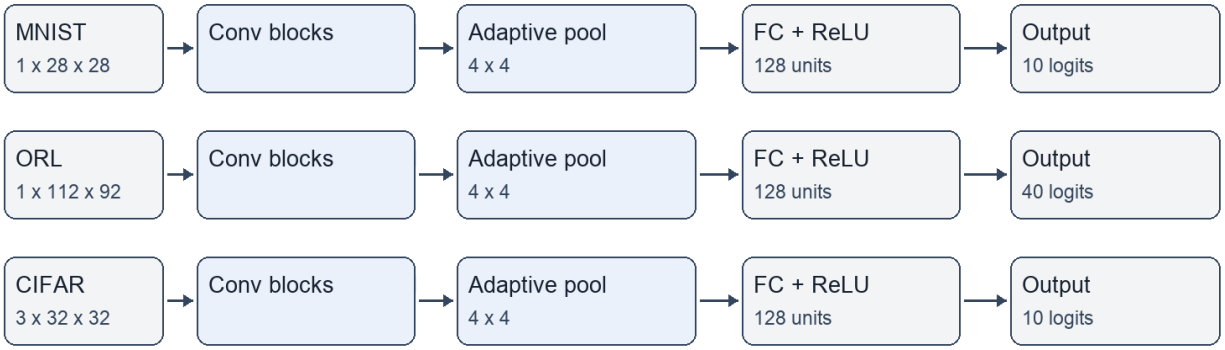


Figure 1. Baseline CNN pipeline. Each convolution block uses ReLU and 2 x 2 max pooling.

Section 3: Experiment Settings

3.1 Dataset Description

MNIST has 60,000 training and 10,000 test images, each 28x28 pixels and labeled 0-9. ORL has 360 training and 40 test face images, each 112x92 pixels and labeled 1-40. Its three stored color channels contain identical grayscale values, so we used one channel. CIFAR-10 has 50,000 training and 10,000 test color images, each 32x32 pixels and labeled 0-9. We reserved a stratified 10% of each supplied training partition for validation and left the test partitions intact. The resulting fit/validation counts were 54,001/5,999 for MNIST, 320/40 for ORL, and 45,000/5,000 for CIFAR. The ORL loader maps labels to 0-39 for training and restores 1-40 in saved predictions. We converted images to channel-first float32 tensors and divided pixel values by 255.

Dataset	Supplied train	Fit	Validation	Test	Input
MNIST	60,000	54,001	5,999	10,000	1 x 28 x 28
ORL	360	320	40	40	1 x 112 x 92
CIFAR-10	50,000	45,000	5,000	10,000	3 x 32 x 32

3.2 Detailed Experimental Setups

All models used Xavier-uniform weights, zero biases, ReLU, cross-entropy loss, and Adam at a learning rate of 0.001. We set the random seed to 42 and used batches of 128 for MNIST and CIFAR and 32 for ORL. MNIST and CIFAR had a ten-epoch cap; ORL had a 35-epoch cap. After each epoch, we saved the checkpoint with the lowest validation loss. Early stopping would have ended training after five epochs without an improvement of at least 0.0001, but all nine runs reached their epoch caps. All runs used the CPU, with no data augmentation. Within each dataset, the variants shared the partition, seed, optimizer, batch size, and epoch cap.

3.3 Evaluation Metrics

Accuracy is the share of test predictions that match the label. We calculate precision, recall, and F1 for each class, then report their unweighted class averages. If precision or recall has a zero denominator, its value is zero. We also give the correct-prediction count for each baseline.

3.4 Source Code

The Stage 3 source code and run instructions are at https://github.com/hiyab-tesfay/ECS-170-Project. The relevant files are local_code/stage_3_code/Dataset_Loader.py, local_code/stage_3_code/Method_CNN.py, script/stage_3_script/script_cnn.py, tests/test_stage3.py, and script/stage_3_script/benchmark_device.py. The repository does not include the dataset files or trained weights. The instructor's dataset is at https://drive.google.com/file/d/1-udd9jkldM1gEQ-OO-9HKXds42AiLOL6/view?usp=sharing.

3.5 Training Convergence Plot

The plots show mean training and validation cross-entropy after each epoch for the baseline and both variants on every dataset. Loss fell substantially across all three datasets. On CIFAR, baseline training loss kept falling after epoch 7 while validation loss fluctuated, which suggests overfitting over the observed ten epochs. The selected validation-loss checkpoints were epochs 10/9/7 for MNIST, 33/33/34 for ORL, and 9/10/10 for CIFAR, ordered as baseline/first variant/second variant.

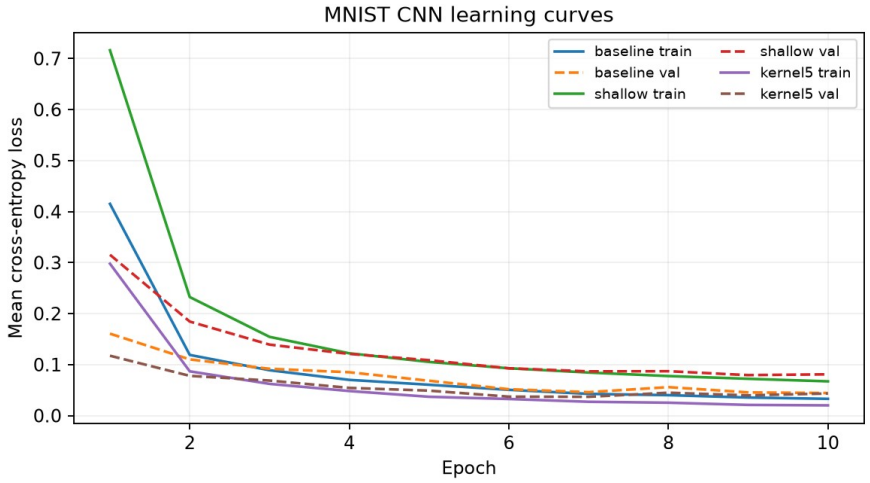


Figure 2a. MNIST training and validation loss.

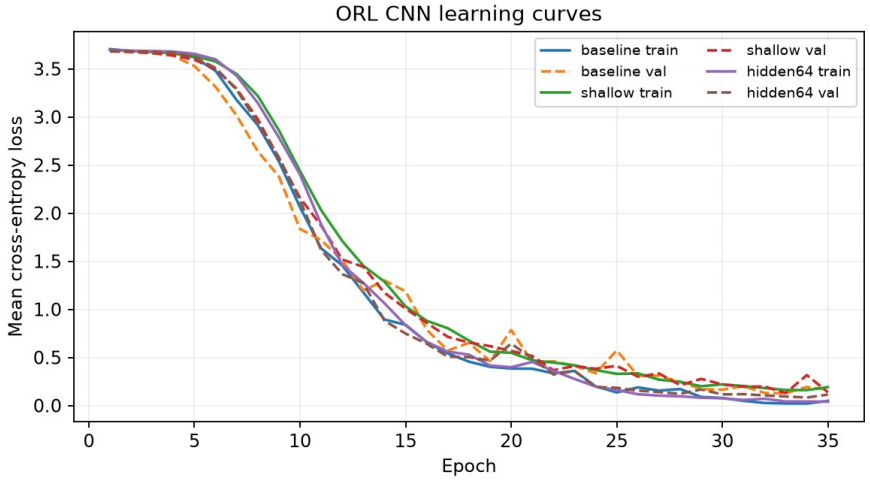


Figure 2b. ORL training and validation loss.

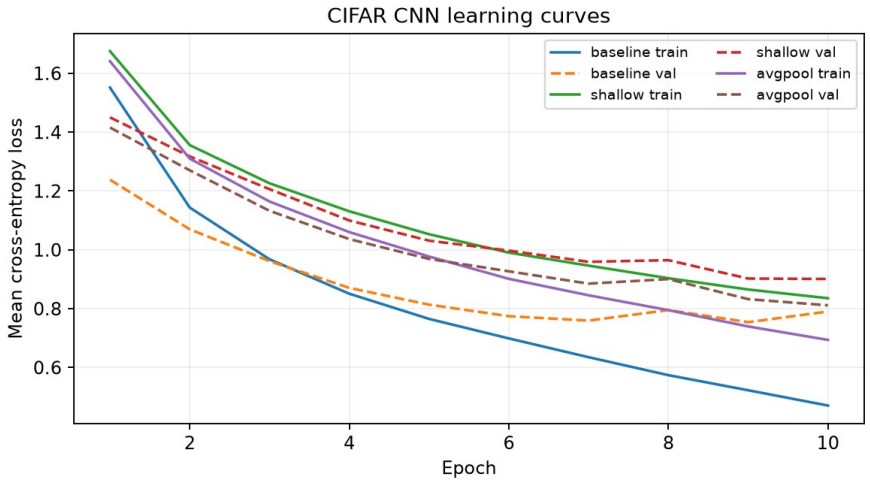


Figure 2c. CIFAR training and validation loss.

3.6 Model Performance

The baseline correctly classified 9,887 of 10,000 MNIST images (98.87%), 39 of 40 ORL faces (97.50%), and 7,410 of 10,000 CIFAR objects (74.10%). The results table reports accuracy and macro precision, recall, and F1 for all nine configurations.

Dataset	Variant	Correct	Accuracy	Precision	Recall	F1
MNIST	Baseline	9,887	98.87%	98.87%	98.86%	98.86%
MNIST	Shallow	9,791	97.91%	97.92%	97.90%	97.90%
MNIST	5 x 5 kernel	9,892	98.92%	98.91%	98.91%	98.91%
ORL	Baseline	39	97.50%	96.25%	97.50%	96.67%
ORL	Shallow	34	85.00%	78.75%	85.00%	80.83%
ORL	64 hidden	37	92.50%	89.58%	92.50%	90.42%
CIFAR	Baseline	7,410	74.10%	74.05%	74.10%	73.95%
CIFAR	Shallow	6,812	68.12%	68.92%	68.12%	67.76%
CIFAR	Avg pool	7,141	71.41%	71.37%	71.41%	71.25%

3.7 Ablation Studies

Removing MNIST's second convolution block lowered test accuracy from 98.87% to 97.91%, or 96 additional errors. Changing its kernels from 3x3 to 5x5 raised accuracy to 98.92%, a gain of five correct images. A single seeded run cannot establish that larger kernels are reliably better. On ORL, the shallower model reached 85.00%, five fewer correct faces than the 97.50% baseline. Reducing the hidden layer from 128 to 64 units reached 92.50%, two fewer correct faces. ORL has one test image per person, so a single error changes accuracy by 2.5 percentage points. On CIFAR, the shallower model reached 68.12%, or 598 fewer correct images than the 74.10% baseline. Replacing max pooling with average pooling reached 71.41%, or 269 fewer correct images. We ran each configuration with one seed and did not measure run-to-run variation.

3.8 Optional GPU Execution

PyTorch 2.2.2 on the machine used for these experiments reports that neither CUDA nor MPS is available (macOS x86_64). We therefore could not run a GPU comparison. To make the optional experiment reproducible on supported hardware, the repository includes script/stage_3_script/benchmark_device.py. It times identical MNIST baseline training batches on CPU and an available CUDA or MPS device, synchronizing the accelerator before recording elapsed time. On this machine, the CPU processed 2,048 training images in 0.8603 seconds (2,380.68 images per second) in one short benchmark. This timing is not a model accuracy result, and no GPU speedup is claimed.

Sources

Instructor dataset and readme: https://drive.google.com/file/d/1-udd9jkldM1gEQ-OO-9HKXds42AiLOL6/view?usp=sharing
Report template: https://docs.google.com/document/d/1fLUhtUe_Rm4x-sL4N-LxBJ57k0enaCEnQNvWwTUkNY0/edit?usp=sharing
Code: https://github.com/hiyab-tesfay/ECS-170-Project
Experiment outputs: local result/stage_3_result/{mnist,orl,cifar}/metrics.json, loss_history.csv, learning_curves.png, and predictions CSV; result/stage_3_result/device_benchmark.json.